

Shazil Shiraj Katchhi

AI & Machine Learning Engineer

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Cottbus, Germany · Indian Nationality



PROFILE

Computer Science & Engineering graduate specialising in Artificial Intelligence and Machine Learning, currently pursuing an M.Sc. in AI at BTU Cottbus-Senftenberg. Research expertise demonstrated through an IEEE-published paper on sign language translation achieving 91% accuracy. Smart India Hackathon finalist with hands-on experience across computer vision, deep learning, full-stack development, and embedded hardware integration. Passionate about building accessible, impactful AI solutions.

EDUCATION

M.Sc. Artificial Intelligence

2025 – present

Brandenburg University of Technology (BTU) Cottbus-Senftenberg, Germany

Coursework: Machine Learning, Deep Learning, Data Analytics, Statistical Methods.

B.E. Computer Science & Engineering (AI & ML)

2021 – 2025

M.H. Saboo Siddik College of Engineering, University of Mumbai, India

CGPA: 7.14 / 10

Higher Secondary Certificate (HSC)

2019 – 2021

Navneet College of Commerce, Arts & Science, Mumbai, India

Score: 89.6%

PROFESSIONAL EXPERIENCE

Data Analyst — Intern

Jun – Oct 2024

Infosys Limited, Bangalore, India

- Performed data extraction, transformation and loading (ETL) using SQL, Python and cloud services (AWS / Azure) to support analytics workflows.
- Designed, developed and maintained dashboards and reports in Power BI and Tableau to deliver real-time business insights.
- Applied query optimisation, indexing and partitioning techniques to improve database performance and reduce processing time.
- Automated data processing, reporting and validation tasks in Python — improving operational efficiency and accuracy.

PUBLICATIONS

Empowering Deaf Communication: A Deep Learning Framework for Sign-to-Speech Translation

Mar 2025

IEEE Xplore — ESCI 2025, AISSMS IoT, Pune, India (Co-author)

- Designed a real-time Indian Sign Language (ISL) to speech/text translation system using Transformer, CNN (ResNet) and LSTM models with MediaPipe for gesture recognition.
- Integrated NLP and Text-to-Speech (TTS) for multilingual output; achieved 91% gesture-detection accuracy.

SELECTED PROJECTS

Smart Glasses for Speech & Hearing Impaired — Smart India Hackathon Finalist (Top 5) Dec 2024 – Mar 2025

- Built wearable ISL translation glasses using Pose Estimation and Gesture Recognition (MediaPipe) combined with ViT, CNN and LSTM models.
- Implemented real-time TTS output with low-latency processing on Raspberry Pi 3 and Raspberry Pi Zero 2W with Adafruit Voice Bonnet.

Code: github.com/mrchikoo/Sign-Language-to-speech-TPP-FPP-

TEXTURA — Prompt-Based Image Generation System

2025

- Engineered a text-to-image pipeline using advanced diffusion-based deep learning models, fine-tuned on custom datasets with a user-facing interface.
- Integrated language-vision models for context-aware, high-realism image generation.

Code: github.com/mrchikoo/Gen_Ai

Object Detection System

Mar 2024

- Developed a real-time object detection model using Python (OpenCV, TensorFlow) and deep learning; optimised for inference performance.

Code: github.com/mrchikoo/Object_detection

Open Chatroom Web Application

2024

- Built a full-stack real-time chatroom using React.js, Firebase (Google Auth) and Node.js, enabling student collaboration and note sharing.

SKILLS & COMPETENCIES

Programming Python (Pandas, NumPy, Scikit-learn, TensorFlow, Keras, PyTorch, OpenCV, MediaPipe), SQL, R, C/C++, JavaScript

AI / ML Computer Vision, Deep Learning, NLP, Gesture Recognition, Diffusion Models, Object Detection, Transformers, LSTMs

Web & IoT React.js, Node.js, Django, HTML/CSS, Firebase, Node-RED, Raspberry Pi, Embedded ML

Tools Git, Jupyter, Figma, Canva

Methods Analytical problem-solving, structured experimentation, communication of technical findings to non-technical audiences

LEADERSHIP & ACTIVITIES

- Chairperson — IEEE MHSSCE Student Chapter (2024 – 2025)
- General Secretary — SACE Committee, MHSSCE (Jan – Jun 2025)
- Social Media Head — IEEE MHSSCE Student Chapter (Mar – Dec 2023)
- Lead, Graphics Team — SACE Committee, MHSSCE (Feb – Apr 2023)

LANGUAGES

English C2 — Proficient (IELTS 8.5)

German A2 — Elementary (working towards B1)

Hindi Native

Marathi Native

Urdu Native